Alert Notification Basics

Series: WFLOW | **Notebook:** 3 of 9 | **Created:** January 2026

Sending Notifications with Workflows

The most common workflow use case is sending alert notifications to Slack, Microsoft Teams, and email. This notebook covers setting up connections, configuring notification tasks, and best practices for effective alerting.

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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS with Platform subscription
Permissions	`automation:workflows:write`, `automation:connections:write`
Prior Knowledge	**WFLOW-01** and **WFLOW-02**
External Access	Slack workspace admin or Teams channel access

1. Notification Architecture

How Workflow Notifications Work

```
Trigger (Davis Problem, Schedule, etc.)
            ↓
    Workflow Executes
            ↓
    Notification Task
            ↓
    Connection (credentials)
            ↓
    External Service (Slack, Teams, Email)
```

Components

Component	Purpose
Connection	Securely stores credentials (tokens, webhooks)
Notification Task	Sends message using connection
Message Template	Dynamic content using Jinja expressions

Supported Notification Channels

Channel	Task Type	Authentication
Slack	`dynatrace.slack:message`	OAuth App or Webhook
Microsoft Teams	`dynatrace.msteams:message`	Incoming Webhook
Email	`dynatrace.email:send`	SMTP or built-in
PagerDuty	`dynatrace.pagerduty:*`	Integration Key
ServiceNow	`dynatrace.servicenow:*`	OAuth or Basic Auth
Jira	`dynatrace.jira:*`	API Token
Custom Webhook	`dynatrace.http:request`	Bearer/Basic/Custom

2. Setting Up Connections

Connections store credentials separately from workflows for security and reusability.

Accessing Connections

1. Open **Settings** (gear icon)
2. Navigate to **Integration** → **Connections**
3. Or use direct URL: `https://<env>/ui/apps/dynatrace.hub/connections`

Creating a Connection

1. Click **+ Connection**
2. Select connection type (Slack, Teams, etc.)
3. Enter credentials
4. Name the connection (e.g., `slack-production-alerts`)
5. Save

Connection Best Practices

Practice	Description
Descriptive names	`slack-prod-oncall` not `slack1`
Environment separation	Different connections for prod/staging
Least privilege	Minimum required scopes
Regular rotation	Update tokens periodically

3. Slack Notifications

Option 1: Slack App (Recommended)

Provides richer features: channels, DMs, reactions, threads.

Setup:
1. Create Slack App at https://api.slack.com/apps
2. Add OAuth scopes: `chat:write`, `chat:write.public`
3. Install to workspace
4. Copy Bot User OAuth Token
5. Create connection in Dynatrace with token

Option 2: Incoming Webhook (Simpler)

Single channel, no advanced features.

Setup:
1. Go to Slack channel settings → Integrations
2. Add **Incoming Webhook**
3. Copy webhook URL
4. Create connection with webhook URL

Basic Slack Message Task

```yaml
name: send_slack_alert
type: dynatrace.slack:message
input:
  connection: slack-production
  channel: "#alerts-production"
  message: |
    :rotating_light: *Problem Detected*

    *Title:* {{ event()["title"] }}
    *Severity:* {{ event()["severity"] }}
    *Started:* {{ event()["start_time"] }}

    &lt;{{ event()["problem_url"] }}|View in Dynatrace&gt;
```

Slack Message with Blocks

For richer formatting:

```yaml
input:
  connection: slack-production
  channel: "#alerts"
  blocks:
    - type: header
      text:
        type: plain_text
        text: "{{ ':red_circle:' if event()['severity'] == 'CRITICAL' else
':large_orange_circle:' }} {{ event()['severity'] }} Alert"
    - type: section
      fields:
        - type: mrkdwn
          text: "*Problem:*\n{{ event()['title'] }}"
        - type: mrkdwn
          text: "*Status:*\n{{ event()['status'] }}"
    - type: section
      fields:
        - type: mrkdwn
          text: "*Started:*\n{{ event()['start_time'] }}"
        - type: mrkdwn
          text: "*ID:*\n{{ event()['display_id'] }}"
    - type: actions
      elements:
        - type: button
          text:
            type: plain_text
            text: "View Problem"
          url: "{{ event()['problem_url'] }}"
```

4. Microsoft Teams Notifications

Setting Up Teams Webhook

1. Open target Teams channel
2. Click **...** → **Connectors** (or **Workflows** in new Teams)
3. Add **Incoming Webhook**
4. Name it (e.g., "Dynatrace Alerts")
5. Copy the webhook URL
6. Create connection in Dynatrace

> **Note:** Microsoft is deprecating Office 365 Connectors. Consider using Power Automate or the newer Workflows connector for future-proofing.

Basic Teams Message Task

```
name: send_teams_alert
type: dynatrace.msteams:message
input:
  connection: teams-production
  message: |
    **Problem Detected**

    **Title:** {{ event()["title"] }}
    **Severity:** {{ event()["severity"] }}
    **Started:** {{ event()["start_time"] }}

    [View in Dynatrace]({{ event()["problem_url"] }})
```

Teams Adaptive Card

For richer formatting:

```yaml
input:
  connection: teams-production
  card:
    type: AdaptiveCard
    $schema: "http://adaptivecards.io/schemas/adaptive-card.json"
    version: "1.4"
    body:
      - type: TextBlock
        text: "{{ event()['severity'] }} Alert"
        size: Large
        weight: Bolder
        color: "{{ 'Attention' if event()['severity'] == 'CRITICAL' else
'Warning' }}"
      - type: FactSet
        facts:
          - title: "Problem"
            value: "{{ event()['title'] }}"
          - title: "Status"
            value: "{{ event()['status'] }}"
          - title: "Started"
            value: "{{ event()['start_time'] }}"
    actions:
      - type: Action.OpenUrl
        title: "View in Dynatrace"
        url: "{{ event()['problem_url'] }}"
```

5. Email Notifications

Email Options

Option	Setup	Best For
Built-in	No setup required	Simple notifications
Custom SMTP	Configure SMTP server	Corporate email servers
SendGrid/SES	API integration	High volume, templates

Basic Email Task

```yaml
name: send_email_alert
type: dynatrace.email:send
input:
  to:
    - "oncall@company.com"
    - "platform-team@company.com"
  subject: "[{{ event()['severity'] }}] {{ event()['title'] }}"
  body: |
    A problem has been detected in your Dynatrace environment.

    Problem Details:
    ----------------
    Title: {{ event()["title"] }}
    Severity: {{ event()["severity"] }}
    Status: {{ event()["status"] }}
    Start Time: {{ event()["start_time"] }}
    Problem ID: {{ event()["display_id"] }}

    Affected Entities:
    {{ event()["affected_entity_ids"] | join(", ") }}

    Root Cause:
    {{ event()["root_cause_entity_id"] }}

    View this problem:
    {{ event()["problem_url"] }}
```

HTML Email

```
input:
  to: ["oncall@company.com"]
  subject: "[{{ event()['severity'] }}] {{ event()['title'] }}"
  contentType: "text/html"
  body: |
    &lt;html&gt;
    &lt;body style="font-family: Arial, sans-serif;"&gt;
      &lt;h2 style="color: {{ '#dc3545' if event()['severity'] == 'CRITICAL' else '#ffc107' }};"&gt;
        {{ event()['severity'] }} Alert
      &lt;/h2&gt;
      &lt;table style="border-collapse: collapse;"&gt;
        &lt;tr&gt;&lt;td&gt;&lt;b&gt;Problem:&lt;/b&gt;&lt;/td&gt;&lt;td&gt;{{ event()['title'] }}&lt;/td&gt;&lt;/tr&gt;
        &lt;tr&gt;&lt;td&gt;&lt;b&gt;Status:&lt;/b&gt;&lt;/td&gt;&lt;td&gt;{{ event()['status'] }}&lt;/td&gt;&lt;/tr&gt;
        &lt;tr&gt;&lt;td&gt;&lt;b&gt;Started:&lt;/b&gt;&lt;/td&gt;&lt;td&gt;{{ event()['start_time'] }}&lt;/td&gt;&lt;/tr&gt;
      &lt;/table&gt;
      &lt;p&gt;&lt;a href="{{ event()['problem_url'] }}"&gt;View in Dynatrace&lt;/a&gt;&lt;/p&gt;
    &lt;/body&gt;
    &lt;/html&gt;
```

6. Message Formatting

Jinja Expression Reference

Expression	Result	Use
`{{ event()["title"] }}`	Problem title	Main content
`{{ event()["severity"] }}`	CRITICAL/HIGH/MEDIUM/LOW	Color coding
`{{ event()["affected_entity_ids"] | join(", ") }}`	Comma-separated list	Show all affected
`{{ event()["start_time"] }}`	ISO timestamp	When started
`{{ event()["problem_url"] }}`	Full URL	Link to problem

Conditional Formatting

```
{% if event()["severity"] == "CRITICAL" %}
:red_circle: CRITICAL ALERT
{% elif event()["severity"] == "HIGH" %}
:large_orange_circle: HIGH ALERT
{% else %}
:large_yellow_circle: {{ event()["severity"] }} ALERT
{% endif %}
```

Severity Emoji Map

Severity	Slack Emoji	Teams Color
CRITICAL	`:red_circle:`	`Attention`
HIGH	`:large_orange_circle:`	`Warning`
MEDIUM	`:large_yellow_circle:`	`Accent`
LOW	`:large_blue_circle:`	`Good`

Inline Severity Mapping

```
{{ {"CRITICAL": ":red_circle:", "HIGH": ":large_orange_circle:", "MEDIUM":
":large_yellow_circle:", "LOW": ":large_blue_circle:"}.get(event()
["severity"], ":white_circle:") }}
```

7. Complete Alert Workflow Example

A production-ready workflow that sends alerts to Slack, Teams, and email.

Workflow Configuration

```yaml
name: production-alert-notifications
description: Send alerts to all channels for production problems

trigger:
  type: davis-problem
  config:
    entityTagsMatch: all
    entityTags:
      - key: env
        value: prod

tasks:
  - name: slack_notification
    type: dynatrace.slack:message
    input:
      connection: slack-production
      channel: "#alerts-production"
      message: |
        {{ {"CRITICAL": ":red_circle:", "HIGH": ":large_orange_circle:",
"MEDIUM": ":large_yellow_circle:"}.get(event()["severity"], ":white_circle:")
}} *{{ event()["severity"] }} Problem*

        *{{ event()["title"] }}*

        • *Status:* {{ event()["status"] }}
        • *Started:* {{ event()["start_time"] }}
        • *ID:* {{ event()["display_id"] }}

        &lt;{{ event()["problem_url"] }}|:mag: View Problem&gt;

  - name: teams_notification
    type: dynatrace.msteams:message
    input:
      connection: teams-production
      message: |
        **{{ event()["severity"] }} Problem**

        **{{ event()["title"] }}**

        - **Status:** {{ event()["status"] }}
        - **Started:** {{ event()["start_time"] }}
        - **ID:** {{ event()["display_id"] }}

        [View Problem]({{ event()["problem_url"] }})
```

```
  - name: email_notification
    type: dynatrace.email:send
    input:
      to: ["platform-oncall@company.com"]
      subject: "[{{ event()['severity'] }}] {{ event()['title'] }}"
      body: |
        A {{ event()["severity"] }} problem has been detected.

        Problem: {{ event()["title"] }}
        Status: {{ event()["status"] }}
        Started: {{ event()["start_time"] }}

        View: {{ event()["problem_url"] }}
```

Task Execution

By default, tasks execute **in parallel**. All three notifications send simultaneously.

Monitor Your Notification Workflows

```
// Notification workflow executions
fetch events, from: now() - 24h
| filter event.type == "automation.workflow.execution"
| filter contains(workflow.name, "alert") or contains(workflow.name,
"notification")
| fields timestamp,
         workflow.name,
         execution.status,
         execution.duration
| sort timestamp desc
| limit 50
```

```
// Notification task success rates
fetch events, from: now() - 7d
| filter event.type == "automation.task.execution"
| filter contains(task.type, "slack") or contains(task.type, "msteams") or
contains(task.type, "email")
| summarize
    total = count(),
    succeeded = countIf(task.status == "SUCCEEDED"),
    failed = countIf(task.status == "FAILED"),
    by:{task.type}
| fieldsAdd success_rate = round(100.0 * succeeded / total, decimals: 2)
| sort total desc
```

```
// Failed notification tasks
fetch events, from: now() - 24h
| filter event.type == "automation.task.execution"
| filter task.status == "FAILED"
| filter contains(task.type, "slack") or contains(task.type, "msteams") or
contains(task.type, "email")
| fields timestamp, workflow.name, task.name, task.type, task.error
| sort timestamp desc
| limit 20
```

Next Steps

Now that you can send basic notifications, learn advanced patterns:

Recommended Path

1. **WFLOW-04: Advanced Notification Routing** - Conditional routing by severity, team, time
2. **WFLOW-05: PagerDuty & ServiceNow** - Incident management integration
3. **WFLOW-06: Custom Notification Templates** - Rich formatting and dynamic content

Key Takeaways

- **Connections** store credentials securely and are reusable
- **Slack** supports both OAuth apps and webhooks
- **Teams** uses incoming webhooks (consider Power Automate for future)
- **Email** can use built-in SMTP or custom servers
- **Jinja expressions** enable dynamic message content
- Tasks execute in **parallel** by default

Summary

In this notebook, you learned:

- Notification architecture: triggers → tasks → connections → external services
- How to set up connections for Slack, Teams, and email
- Basic and advanced Slack message formatting
- Microsoft Teams notifications with Adaptive Cards
- Plain text and HTML email notifications
- Jinja expressions for dynamic content
- A complete multi-channel alert workflow

References

- Workflow Notifications
- Slack Actions
- Microsoft Teams Actions
- Email Actions
- Jinja Expressions

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